When AI Is Not Optional: Structural Constraint and Participation in Intercultural Healthcare Communication

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Abstract

In this position paper, we argue that AI-mediated communication in intercultural healthcare cannot be understood as merely an optional technological enhancement. Prior work often frames AI systems as tools that improve efficiency, usability, and user control, implicitly assuming that adoption is voluntary. However, in contexts where institutional language support is limited, AI functions less as a facilitative aid and more as a substitute infrastructure for communication. Drawing on analyses of intercultural medical communication in South Korea, we contend that AI use in such settings is frequently compelled rather than chosen, redistributes responsibility and risk asymmetrically, and renders participation procedural rather than substantively participatory. When mediation becomes infrastructural, evaluating AI solely by translation quality or usability fails to capture its relational consequences. We propose reframing authenticity in AI-mediated healthcare communication as the capacity for meaningful participation under structural constraint and invite reconsideration of how such systems are designed and assessed in high-stakes intercultural settings.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

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1 Introduction

AI-mediated communication is frequently framed as a facilitative enhancement that improves interaction, decision-making, and efficiency, with research largely centered on optimizing usefulness, trust, and user control[3, 4, 8, 11]. By emphasizing user control and preference-sensitive adoption, this body of work implicitly assumes that individuals can meaningfully opt in or opt out of AI systems. However, this assumption does not hold across all communication contexts.

In healthcare settings, AI-mediated communication operates within interactions that are already characterized by asymmetries of knowledge, authority, and responsibility between clinicians and patients[5, 6, 21, 23, 25]. These encounters are inherently complex, as they involve not only the exchange of information but also uncertainty, vulnerability, and high-stakes decision-making. In intercultural healthcare contexts, this complexity is further intensified by differences in cultural norms and expectations that shape how patients express concerns and respond to medical authority[10, 18, 22]. Communication failures can have serious consequences, alternatives such as professional interpretation are often limited, and opting out of mediated communication is rarely a realistic option. For foreign patients navigating medical encounters, AI-based translation and communication tools frequently become the only viable means of interaction. Under these conditions, AI use is not a matter of preference but a response to structural constraints that shape how communication can take place.

In this position paper, we examine AI-mediated communication under constrained conditions in intercultural healthcare, using the South Korean healthcare system as a case. We focus on situations in which foreign patients rely on AI in order to meaningfully participate in medical decision-making. Within this context, restoring human authenticity is understood not as making AI-generated language sound more natural, but as enabling genuine participation in medical encounters. By presenting this case and reflecting on ongoing qualitative work with foreign residents, we offer a conceptual perspective on AI-mediated communication as a structurally constrained practice. Rather than evaluating AI solely in terms of accuracy or usability, we consider how it reshapes participation, responsibility, and relational dynamics in high-stakes intercultural settings. AI-mediated communication is therefore not merely a technical tool for transferring information, but a relational mediator that reorganizes how patients and clinicians engage under structural constraint. This perspective invites reconsideration of how such systems are conceptualized, designed, and evaluated.

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2 Conceptual Background: Participation, Asymmetry, and Cultural Context

2.1 Medical Communication and Participation

Medical communication has long been understood as more than the transmission of clinical information[21]. Research on shared decision-making emphasizes that patients are not passive recipients of medical expertise but participants whose values, preferences, and concerns should inform diagnostic and treatment decisions[6, 7, 21]. Informed consent, likewise, is not merely a formal signature on a document but a communicative process through which patients come to understand risks, alternatives, and implications of care.

At the same time, participation unfolds within structurally asymmetrical relationships[5, 25]. Clinical encounters are organized by differences in institutional authority, professional knowledge, and control over procedures and time. clinicians occupy positions of recognized expertise, while patients often navigate uncertainty and vulnerability. Even in linguistically shared contexts, patients may hesitate to request clarification or challenge recommendations. Participation, therefore, cannot be equated with the mere presence of information exchange; it depends on whether patients are able to articulate concerns, negotiate meanings, and meaningfully influence decisions within unequal relational conditions.

2.2 Culture as a Condition of Participation

Culture is a central factor shaping health communication and behavior[2, 15]. Beyond language, culture reflects values, norms, beliefs, and communicative practices that structure how health problems are defined, how risks are interpreted, and how authority is negotiated[10, 18]. Cultural contexts influence health priorities, decision-making patterns, and expectations about appropriate roles in clinical encounters[22].

In this sense, culture functions not simply as a background variable but as a condition of meaning-making. It shapes what can be expressed, how uncertainty is managed, and whether questioning authority is considered appropriate. For example, collectivistic orientations have been associated with stronger in-group boundaries and differentiated expectations toward authority figures[14]. In contexts where deference, relational harmony, or implicit communication norms are emphasized, patients may be less likely to openly express doubt or disagreement, even when confusion persists.

When layered onto already asymmetrical medical relationships, such culturally informed norms can further constrain participation. Intercultural healthcare encounters therefore intensify preexisting asymmetries by introducing differences in expectations about authority, responsibility, and communicative legitimacy. Understanding culture in this way reframes participation not as access to translated information alone, but as access to the relational and normative conditions that make questioning, clarification, and negotiation possible.

3 Intercultural Medical Communication in South Korea

South Korea has experienced a steady rise in its foreign resident population since the 1990s, driven by increasing numbers of migrant workers and multicultural families[13]. As of 2024, approximately 2.65 million foreign residents were living in Korea, accounting for 5.2% of the total population[16]. Despite this demographic shift, most medical institutions continue to operate primarily in Korean, with limited systematic provision of professional interpretation or multilingual support, reflecting persistent gaps in intercultural preparedness as healthcare systems adjust to increasing multicultural diversity[13, 20]. These structural gaps have become more pronounced in the post-COVID era. The pandemic accelerated the digital transformation of healthcare services and expanded reliance on AI-based communication tools[1, 19, 26, 27]. Under these conditions, pre-existing institutional and relational limitations have become increasingly visible. As a result, communication in clinical settings often depends not on institutional infrastructure, but on informal solutions such as accompanying spouses, friends, or ad hoc digital translation tools.

Prior research has documented the consequences of these structural limitations. Studies report that many foreign patients experience dissatisfaction and anxiety during medical encounters due to linguistic barriers, limited access to health information, and difficulties navigating the Korean healthcare system[12, 17, 24]. In some cases, language barriers have led to serious ethical and legal concerns, including procedures performed without fully informed consent and misunderstandings regarding treatment decisions shaped by religious or cultural considerations[13]. These findings highlight that the challenge is not merely one of individual language proficiency but of institutional preparedness and intercultural competence.

Importantly, reliance on informal mediators introduces additional tensions. Several studies note that foreign patients often depend on Korean spouses or acquaintances to interpret medical information[9], raising concerns about privacy, autonomy, and the accurate conveyance of sensitive health details. In such settings, meaningful participation in decision-making becomes conditional on external assistance rather than institutional support.

Our preliminary interviews with foreign residents who had experienced medical encounters in South Korea echo these structural patterns. Participants described relying on AI tools not because they fully trusted them, but because they had no alternative forms of language support. As one participant stated, *"It's not about the trust… It's about why we don't have an option."* Others described situations in which attempts to use phone-based translation were met with frustration from medical staff, leading them to call friends or spouses for assistance instead. Several participants emphasized that they felt more comfortable attending appointments with someone who could help interpret, highlighting how participation often depended on informal support rather than institutional infrastructure. These accounts suggest that AI and other informal mediation tools function less as optional enhancements and more as necessary intermediaries in structurally constrained encounters. While communication may proceed at a procedural level, patients' ability to question, clarify, or negotiate decisions remains shaped by the limited conditions under which interaction takes place.

4 AI-Mediated Communication as Structural Constraint: Arguments and Implications

Building on the structural and cultural conditions outlined above, we argue that AI-mediated communication in intercultural healthcare should be understood as operating within constrained conditions that reconfigure participation, responsibility, and authenticity in medical encounters. We advance three related arguments:

- **AI use is compelled rather than voluntary.** In contexts where institutional language support is limited, foreign patients rely on AI not out of preference, but necessity. Opting out is often unrealistic.
- **Responsibility and risk are unevenly redistributed.** Translation errors and misunderstandings are frequently absorbed by patients, who lack the authority or linguistic capacity to verify or contest mediation.
- **Participation becomes procedural rather than substantive.** While communication formally occurs and consent is obtained, the conditions for questioning, clarification, and negotiation remain fragile.

These arguments challenge prevailing framings of AI-mediated medical communication that prioritize accuracy, usability, or trust as primary evaluation criteria. Yet when mediation becomes structurally unavoidable, such performance-oriented metrics are insufficient to account for its relational and participatory consequences. Accordingly, evaluation and design approaches must be reoriented to address these broader relational and structural implications.

First, AI systems in high-stakes intercultural settings should be evaluated not only by translation quality but also by whether they enable substantive participation in decision-making. Second, design approaches must account for asymmetrical responsibility structures, ensuring that errors do not disproportionately burden patients. Third, authenticity in AI-mediated communication should be reconsidered as a relational and participatory condition rather than a linguistic one.

We invite discussion on the following questions:

- How should AI-mediated communication be designed in contexts where mediation is structurally unavoidable?
- What recurring risks emerge when AI becomes infrastructural in intercultural healthcare communication, and where might meaningful opportunities lie?
- How can AI-mediated systems in high-stakes intercultural settings support autonomy and authentic participation without masking structural asymmetries?

Addressing these questions requires reconsidering not only how AI systems perform, but how they reshape the conditions under which meaningful participation becomes possible.

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